Python Programming - 2301CS404

Lab - 3

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01) WAP to check whether the given number is positive or negative.

In [3]:
```
num = int(input("Enter Number:"))

if num > 0:
          print("Positive Number")
else:
        print("Negative Number")
```

```
Negative Number
```

02) WAP to check whether the given number is odd or even.

In [4]:
```
num = int(input("Enter Number:"))

if num % 2 == 0:
          print("Even Number")
else:
        print("Odd Number")
```

```
Even Number
```

03) WAP to find out largest number from given two numbers using simple if and ternary operator.

In [9]:
```
num1 = int(input("Enter Num1:"))
num2 = int(input("Enter Num2:"))
# if num1 > num2 :
#     print(f"{num1} is greater")
# else:
#     print(f"{num2} is greater")
```

```
result = print(f"{num1} is greater") if num1 > num2 else print(f"{num2} is great
```

```
6 is greater
```

04) WAP to find out largest number from given three numbers.

In [14]:
```
a = int(input("Enter a:"))
b = int(input("Enter b:"))
c = int(input("Enter c:"))
# First Way

# if a > b and a > c :
#     print(f"{a} is greater")
# elif b > a and b > c:
#     print(f"{b} is greater")
# else:
#     print(f"{c} is greater")


# Second Way
max = a
if b > max:
    max = b
if c > max:
    max = c
print(f"{max} is greater")
```

```
6 is greater
```

05) WAP to check whether the given year is leap year or not.

[If a year can be divisible by 4 but not divisible by 100 then it is leap year but if it is divisible by 400 then it is leap year]

In [19]:
```
year = int(input("Enter year:"))

if (year%4==0 and  year%100!=0) or (year%400==0):
    print(f"{year} is a Leap Year")
else:
    print(f"{year} is not a Leap Year")
```

```
2024 is a Leap Year
```

06) WAP in python to display the name of the day according to the number given by the user.

In [25]:
```
day = int(input("Enter day:"))

match day:
    case 1: print("Sunday")
    case 2: print("Monday")
    case 3: print("Tuesday")
    case 4: print("Wednesday")
    case 5: print("Thursday")
    case 6: print("Friday")
```

```
    case 7: print("Saturday")
    case _ : print("Invalid Choice")
```

```
Friday
```

07) WAP to implement simple calculator which performs (add,sub,mul,div) of two no. based on user input.

In [27]:
```
a = int(input("Enter a:"))
b = int(input("Enter b:"))
Operation = input("Enter Operation(+,-,*,/):")

match Operation:
    case '+':
        print(f"{a} + {b} = {a+b}")
    case '-':
            print(f"{a} - {b} = {a-b}")
    case '*':
            print(f"{a} * {b} = {a*b}")
    case '/':
            print(f"{a} / {b} = {a/b}")
    case _:
            print("Invalid Choice")
```

```
4 + 5 = 9
```

08) WAP to read marks of five subjects. Calculate percentage and print class accordingly.

Fail below 35

Pass Class between 35 to 45

Second Class

between 45 to 60

First Class between 60 to 70

Distinction if more than 70

In [38]:
```
m1,m2,m3,m4,m5 = input("Enter m1,m2,m3,m4,m5:").split(" ")
m1 = int(m1)
m2 = int(m2)
m3 = int(m3)
m4 = int(m4)
m5 = int(m5)
per = (m1+m2+m3+m4+m5)/5
print(f"Percentage: {per}")

if per > 70:
    print("Distinction")
elif per > 60 and per < 70:
    print("First Class")
elif per > 45 and per < 60 :
    print("Second Class")
elif per > 35 and per < 45 :
```

```
    print("Pass Class")
else:
    print("Fail")
```

```
Percentage: 32.4
Fail
```

09) WAP to find the second largest number among three user input numbers.

In [40]:
```
a = int(input("Enter a:"))
b = int(input("Enter b:"))
c = int(input("Enter c:"))

if (a > b and a < c) or (a < b and a > c) :
        second = a
elif (b > a and b < c) or (b < a and b > c) :
        second = b
else :
    second = c
print(f"Second Largest is :{second}")
```

```
Second Largest is :9
```

10) WAP to calculate electricity bill based on following criteria. Which takes the unit from the user.

a. First 1 to 50 units – Rs. 2.60/unit

b. Next 50 to 100 units – Rs. 3.25/unit

c. Next 100 to 200 units – Rs. 5.26/unit

d. above 200 units – Rs. 8.45/unit

In [43]:
```
unit = int(input("Enter Units:"))
if unit >= 200:
    ans = (unit - 200)*8.45
    ans += 50*2.60 + 50*3.25 + 100*5.26
elif unit >= 100 and unit <= 200:
    ans = (unit - 100)*5.26
    ans += 50*2.60 + 50*3.25
elif unit >= 50 and unit <= 100:
    ans = (unit - 50)*3.25
    ans += 50*2.60
elif unit >= 1 and unit <= 50:
    ans  = unit*2.60
print(f"Electricity Bill:{ans}")
```

```
Electricity Bill:195.0
```

Thank You